

# The Erdős–Straus workbench

## a reader of its results, with proofs

An audit and reading prepared by Claude (Anthropic),  
model `claude-opus-5-5` (Opus 5.5), at maximum reasoning effort

Workbench: The Clankers (KokunoYumeto and the contributors credited below)

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### Abstract

The Erdős–Straus workbench of The Clankers (KokunoYumeto/erdos-strauss-foundation and the Zenodo collection 10.5281/zenodo.20401937) studies the equation  $4/n = 1/x + 1/y + 1/z$ . Its main documents run to more than a thousand pages:

- a 619-page cumulative archive (31 August 2026) and an earlier 488-page reader (7 August);
- a 67-page and an 86-page supplement on bounded congruence transport (8 and 9 September);
- a 124-page reader of 20 September;
- a continuation of 17–23 September with 28 structural items.

This reader says what these documents establish, where each result is proved, and how far each was checked here. Its core (Sections 3–4) is proved in full here: the divisor criterion for a fixed first denominator, the halved range of the least denominator, the first two residual shells and their survivors, the classical reduction modulo 840 by five identities, the Jacobi-symbol barrier, the least-class function, and four explicit families that solve the equation for all  $n$  in four arithmetic progressions, each meeting every residue class modulo 840 that the classical reduction leaves open. The record primes of the least-class function and the finite data of the workbench’s record-shell conjecture are recomputed. Section 2 maps the rest: every section of the 619-page archive with its page range and its bearing on the equation, and the supplements and the continuation. It also shows that one theorem of the supplement, on shear transport between adjacent shells, holds under a weaker hypothesis than stated, and that this hypothesis cannot be weakened further. Section 6 states the negative results with their exact scope. The workbench does not claim a proof of the conjecture, and nothing here proves or disproves it.

## About this reader

**The workbench, and who made it.** The collective author name of the workbench is *The Clankers*. The workbench credits:

- `u/CommonCareful3149`, for the mod-107 arithmetic-progression observation;
- `u/UmbrellaCorp_HR`, for arithmetic, positivity-code and prime-production contributions and for the central-angle and inversive-circle theorems of the archive’s §6;
- `u/CivQ17`, for the diagrams behind result R05;
- `u/Far-Lifeguard519`, for the initiating computational investigation;
- `u/TextBackground496` and `u/JGPTech`, for source ideas;
- the owner, KokunoYumeto (`u/lepthymo`), for the research directions and examples, for directing and developing the investigations with AI, and for the proof and source-fidelity requirements.

These credits are from `README.md`, “Attribution and collaboration”, and `CONTRIBUTIONS.md`. The archive records language and computational models as contributing-model provenance for particular calculations, not as authors (archive, “Research status and provenance”).

**What was read.**

- The repository at commit d20b32e (23 September 2026), the latest on 26 September.
- The 488-page reader and its source, which are in the repository.
- The 619-page archive (00\_ERDOS\_STRAUSS\_Project\_Reader.pdf) and its source archive (01\_...\_Source\_Verification\_and\_Machine\_Memory.zip), downloaded from the Zenodo record 10.5281/zenodo.22678971.

Locators have the form “archive § $n$ , p.  $m$ ” for the 619-page PDF, “supplement § $n$ ” for the 86-page supplement, and file names and line numbers for the repository. The audit notes cited as 21\_, 43\_, 43a\_, 47\_ and 47a\_ are on the branch `claude/claude-ab-grind-20260925` of `KokunoYumeto/zeta-function-research-reader`, in `contrib/claude-ab/grind_20260925/`.

**How it was checked.**

- The core of Sections 3–4 was re-derived by the author of this reader. Most of it was refereed three times as part of an earlier reader of four workbenches [9] (Appendix B).
- The new computations of this reader were made with its own scripts: the record primes of the least-class function, the deficit sets of the record shells, the four families, and the exact scope of one theorem of the supplement (Appendix B).
- Everything else carries a status label saying how far it was checked.

This version has not yet had an independent referee pass of its own. Appendix B says why, and what is planned.

**Status labels.**

- Proved here: the proof is given in full in this reader.
- Audited: the workbench’s proof was read in full, every step was checked, and independent checks were run.
- Audited (certificate): the finite certificate on which a proof rests was recomputed independently, and the deduction from it was read.
- Audited in part: the parts named were checked; the rest is located.
- Reproduced: the finite content of a statement (its examples, tables, identities or counts on a stated range) was recomputed by an independent script; its general proof was not re-derived.
- Located: the statement and its proof were found and are described; nothing was checked.

The workbench itself labels statements as proved, certificate-checked, conjectural or withdrawn. Its label is quoted where it differs from the one above.

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# 1 The problem, what is known, and what the workbench claims

## 1.1 The equation

The Erdős–Straus conjecture asserts that for every integer  $n \geq 2$  there are positive integers  $x, y, z$  with

$$\frac{4}{n} = \frac{1}{x} + \frac{1}{y} + \frac{1}{z}.$$

Repeated denominators are allowed; for example  $4/5 = 1/2 + 1/4 + 1/20$ . It suffices to treat primes:  $4/2 = 1/1 + 1/2 + 1/2$ , and if  $4/p = 1/x + 1/y + 1/z$  then  $4/(mp) = 1/(mx) + 1/(my) + 1/(mz)$ .

## 1.2 What is known

The following results frame everything below. Each is cited from the literature; none is re-proved here, except where Section 3 proves a special case.

- *The classes modulo 840.* Explicit polynomial identities solve the equation for every  $n$  in a primitive class  $r \bmod 840$  unless  $r$  is a square modulo 840. So for primes only the classes

$$p \equiv 1, 121, 169, 289, 361, 529 \pmod{840}$$

remain. Elsholtz and Tao attribute this to Obláth and to Mordell [2, 5]; Proposition 3.5 below proves the reduction for primes.

- *Why identities cannot finish the job.* A primitive class  $n \equiv r \pmod{q}$  with  $r$  a square modulo  $q$  cannot be solved by polynomials; Elsholtz and Tao trace this observation to Schinzel and to Yamamoto [2]. Section 3.4 gives the workbench’s shell form of this obstruction, with proof.
- *Density.* The number of  $n < N$  for which the equation has no solution is at most  $N \exp(-c \log^{2/3} N)$  for an absolute  $c > 0$  and  $N$  large (Vaughan [7], as stated by Elsholtz and Tao [2]).
- *Counting.* Elsholtz and Tao bound the number  $f(n)$  of solutions and its averages, and classify solutions at primes as Type I ( $p$  divides exactly one denominator) or Type II ( $p$  divides two) [2].
- *Verification.* The conjecture has been checked for all  $n \leq 10^{14}$  (Swett, cited in [2]) and for all  $n \leq 10^{17}$  (Salez [6]); a 2025 preprint reports the bound  $10^{18}$  [4].

## 1.3 What the workbench claims

The repository’s README opens by saying that the project claims neither a proof nor a counterexample. The research state calls it an ongoing programme, not a completed proof. The archive’s status section (p. 4) names the remaining global obligation: for every relevant prime, at least one marked shell class must be shown to be occupied. The workbench’s scope statements are careful and, in the parts read here, accurate. Where a workbench statement was found to be wrong or overstated, this reader says so and gives the correction (Section 6 and Appendix A).

## 1.4 The approach in one paragraph

Fix a prime  $p$  and a candidate least denominator  $a$  with  $p/4 < a < p$ . The remaining equation  $1/y + 1/z = 4/p - 1/a$  has solutions exactly when  $a^2$  has a divisor in one of two prescribed residue classes modulo  $R = 4a - p$  (Theorem 3.1). The workbench calls  $a$  a *shell* and  $R$  its *residual*. For  $p \equiv 1 \pmod{4}$  the residual is  $\equiv 3 \pmod{4}$ , and the shells in play are  $R = 3, 7, 11, \dots$  up to about  $p$ . Most of the workbench’s arithmetic is organised around three kinds of tool:

- sufficient conditions under which a shell is occupied, stated as congruence conditions on  $p$  or on the prime factors of  $a$ ;

- obstructions under which a shell is empty;
- bounded “carrier” censuses, which count how much of a finite space of residue data these conditions cover.

Around this arithmetic core the documents also develop geometric and operator-theoretic constructions (inversive circles, Lorentz lifts, Boolean carriers, an auxiliary torus, a Busy-Beaver map). Section 2 places each of them and says how it bears on the equation.

## 2 The documents, and what each contains

### 2.1 The editions and how they relate

| Edition                                             | Where                                                                  | Contents and relation to the others                                                                                                                                                                                                                                                                                                        |
|-----------------------------------------------------|------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Project reader, 488 pages, 7 August 2026            | repository root, with source and Lean archives                         | The earlier edition. The README says it is preserved and is not the newest reader.                                                                                                                                                                                                                                                         |
| Cumulative archive, 619 pages, 31 August 2026       | Zenodo 22666493 and 22678971, file 00                                  | The 488-page reader continued. Its main source has the same section skeleton, now 60,561 lines instead of 57,452. Fourteen satellite files add §§10–19 and the carrier refinements §§24.1–24.3. It begins from the source of the earlier Zenodo record 21845035, version 69 (archive p. 3). It is the front file of the Zenodo collection. |
| Bounded congruence transport, 67 pages, 8 September | Zenodo 22666493, files 02–03                                           | The 58-statement supplement.                                                                                                                                                                                                                                                                                                               |
| Expanded supplement, 86 pages, 9 September          | Zenodo 22678971, files 07–08; repository research/expanded-2026-09-08/ | The same, with 14 further statements: 72 statements in 15 source files, with checkers (§2.3).                                                                                                                                                                                                                                              |
| ES/Fable reader, 124 pages, 20 September            | repository research/incoming/es-fable-zeta-bridge-20260920/            | A coordinate bridge from Erdős–Straus solutions to the Fable cover and a weighted conductor of the zeta programme (§2.4).                                                                                                                                                                                                                  |
| Continuation, 17–23 September                       | repository README, “Start reading” (28 items); research/incoming/      | 41 dated packages; the results bench R01–R10 and the integration arguments X1–X7 (§2.5).                                                                                                                                                                                                                                                   |

These editions have different contents; the repository preserves them rather than overwriting them (README, edition table).

### 2.2 The 619-page archive, section by section

Page numbers are those of the PDF. The column “Bearing” says whether a section bears on the equation:

- *direct*: it proves or tests solvability;
- *literature*: it audits a published approach to the equation;
- *side*: a construction not about solvability.

The status column refers to this reader.

| §     | pp.     | Content                                                                                                                     | Bearing                              | Status here                                                             |
|-------|---------|-----------------------------------------------------------------------------------------------------------------------------|--------------------------------------|-------------------------------------------------------------------------|
| front | 3–5     | Research status: the census figures of §24, the four families, the record-shell results, and an explicit list of non-claims | summary                              | read                                                                    |
| 1     | 6–16    | Scope, road map, mathematical status, attribution, lineage                                                                  | summary                              | read in part                                                            |
| 2     | 16–18   | The discriminant-five identity $(2n+1)^2 - 5 = 4(n^2 + n - 1)$ and its root involution, from u/CivQ17                       | side                                 | located                                                                 |
| 3     | 18–20   | The shell $p_* = 8,803,369$ , $R_* = 107$ , reached through Busy-Beaver values ( $S(4) = 107$ , $\text{Space}(5) = 12289$ ) | direct (one shell)                   | $R = 107$ is the least occupied shell at $p_*$ (Proposition 4.2) see §7 |
| 4     | 20–22   | Alpöge’s constant-Jacobian map $F : \mathbb{C}^3 \rightarrow \mathbb{C}^3$ with a three-point fibre                         | side                                 |                                                                         |
| 5     | 22      | Withdrawn constructions (a cube-root interpolation on the Fable fibre), with the point of failure                           | side                                 | read                                                                    |
| 6     | 22–27   | Central-angle and inversive-circle theorems of u/UmbrellaCorp_HR; a Lorentz reflection                                      | side                                 | located                                                                 |
| 7     | 27–31   | The complete fibre of $F$ over $(-\frac{1}{4}, 0, 0)$ and a marked-factor lift                                              | side                                 | see §7                                                                  |
| 8     | 31–38   | The singular $C_1$ – $C_2$ – $C_3$ support object; dihedral actions                                                         | side                                 | located                                                                 |
| 9     | 38–93   | The exact divisor-shell bijection, then directed blade transport and a quarter-radius inversion                             | direct (§9 opening), side (the rest) | bijection proved here (Theorem 3.1)                                     |
| 10    | 93–98   | Residual sieve modulo 840 and the Jacobi-symbol barrier                                                                     | direct                               | proved here (Propositions 3.5, 3.6)                                     |
| 11    | 98–100  | Elsholtz–Tao Type I and Type II coordinates mapped to the residual shell                                                    | literature                           | located                                                                 |
| 12    | 100–106 | Adjacent programmes (López, Dyachenko, Bradford): what each proves, and where a claimed bridge fails                        | literature                           | located                                                                 |
| 13    | 106–114 | Mordell’s constructions and Rosati’s theorem; Salez’s seven charts; Bright–Loughran surfaces; the Vaughan boundary          | literature                           | located                                                                 |
| 14    | 114–119 | López’s structural solutions: exact divisor coordinates and a quantifier repair                                             | literature                           | located                                                                 |
| 15    | 119–124 | The Star–Kneser programme: a one-way forcing theorem and a retracted converse                                               | direct                               | located                                                                 |
| 16    | 124–127 | The mod-107 deficit progression, following u/CommonCareful3149                                                              | direct                               | progression and its corollary reproduced (§4.1)                         |
| 17    | 127–135 | Two-fraction criteria; Monks–Velingker; Ionascu–Wilson classes; Zelator’s prime-square classification                       | literature                           | classes replayed (§4.1)                                                 |
| 18    | 135–148 | The residual-eleven support ideal, its colons, and the paired $R = 7/R = 11$ affine theorem                                 | direct                               | located                                                                 |
| 19    | 148–157 | Strict record shells: an open conjecture and the boundary example $p = 806,521$                                             | direct                               | records and deficits reproduced (§4.1)                                  |

| §  | pp.     | Content                                                                                                                                                                                                                                                                      | Bearing                               | Status here                                                                                                        |
|----|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| 20 | 157–179 | The shell $p = 12289$ , $R = 11$ , $a = 3075$ : counts $(3, 6, 3)$ , its dihedral orbit, and the Star–Kneser quotient certificate                                                                                                                                            | direct (one shell)                    | counts reproduced: they are $( E_a ,  M_a ,  E_a )$ , and $a$ is the least occupied shell; the rest located see §7 |
| 21 | 179–200 | The split-zero semiring $G(R) = R \sqcup \{\tau\}$ and the $C_1$ carrier                                                                                                                                                                                                     | side                                  |                                                                                                                    |
| 22 | 200–201 | Rigidified denominator towers over $\mathbb{F}_1$                                                                                                                                                                                                                            | side                                  | located                                                                                                            |
| 23 | 201–250 | Mirror completion of the Boolean carrier; the 12289 orbit matrix                                                                                                                                                                                                             | side                                  | located                                                                                                            |
| 24 | 250–604 | Tomita–Takesaki transport and $C_3$ modular data (opening part); then the carrier census: the $R = 167$ and $R = 239$ carrier theorems, the refresh of the $R = 167$ coefficient, and the return to the $R = 23$ lineage (pp. 451–604), including the four families (p. 423) | direct (census), side (operator part) | four families proved here (Theorem 4.3); census located (§4.3)                                                     |
| 25 | 605–616 | A map from Busy–Beaver extrema to finite shell carriers                                                                                                                                                                                                                      | side                                  | see §7                                                                                                             |

So about 100 of the 619 pages bear directly on the equation: §§3, 9 (opening), 10, 15, 16 and 18–20, and the census part of §24. The literature audits of §§11–14 and 17 add about 35 pages. The rest is side constructions, whose connection to the equation the archive itself describes as an interface still to be supplied by an explicit typed map (archive p. 4).

### 2.3 The bounded-transport supplement (72 statements)

The supplement starts from the primes  $p = 12h + 1$  and all shells  $3h + 1 \leq a \leq 9h$  of Theorem 3.1. Its sections are:

1. original shells, coordinates and integer reconstruction, with the first-two-shell sieve and an all-shell no-hit sieve;
2. local congruences with all integral lifts retained, with an integral cyclic bridge;
3. bounded divisibility and simultaneous congruences: unary Boolean transport, a shared-variable Chinese-remainder construction credited to a colleague, raw-scale hit incidence, and a boundary-swap completion;
4. exact cross-shell transport at a fixed rational ratio;
5. a primitive Euclidean successor with its full arithmetic domain;
6. arithmetic localisation along the original norm subdivision, with a reading of Connes’ primitive intersections;
7. a strictly decreasing denominator with exact arithmetic return;
8. the Navier–Stokes auxiliary cover: the torus matrix  $J = \begin{pmatrix} 3 & 1 \\ 1 & 5 \end{pmatrix}$  of the announced Navier–Stokes manuscript, its covering fibres and character corrections, and sampling and reconstruction estimates.

The supplement’s own README states that its results prove or disprove neither the Erdős–Straus conjecture nor the Riemann hypothesis nor the Yang–Mills mass gap, and that they do not validate the Navier–Stokes construction.

Status. Reproduced in part. A second-pass audit, interrupted by a usage limit before it wrote its report, recomputed the finite content of the statements of Sections 1–7. Its scripts, which are independent of the workbench’s code, found every checked statement correct:

- the gluing and Chinese-remainder theorems on random systems and on all shells of small primes;
- the ratio-transport domain on all 57,882 ordered shell pairs of the primes  $p \equiv 1 \pmod{12}$  below 260;
- the shear and no-two-shears theorems on all shells of 131 primes below 4000;
- the Connes marked successor on all primes  $p \equiv 1 \pmod{12}$  below  $3 \cdot 10^5$ ;
- the fixed- $y$  successor on all 3,114 source witnesses below 1500;
- the orchard partition on all 42 admissible triples below 1300;
- the boundary four-candidate law on all 72,282 raw records below 400;
- the unary box, the input/output incidence and the raw-scale theorems on their stated ranges.

One of its checks, on the smallest member of a boundary orbit, reported mismatches. The script expected the pattern  $(\min(A, B), \max(A, B))$  in every case. The supplement’s own table (boundary\_swap\_completion.tex, lines 296–309) gives a one-point orbit’s own point as its smallest member, which is what the script found. The mismatch is therefore in the script, not in the supplement. The same pass also looked at scope, and its search led to Proposition 2.1 below. The general proofs were not re-derived, except that of Section 5’s no-two-shears theorem; the auxiliary-torus Section 8 was not checked. The scripts are in checks/es\_reader/second\_pass\_partial/.

Section 5 of the supplement studies the two unimodular shears  $L(m, n) = (m + n, n)$  and  $J(m, n) = (m, m + n)$  as maps between adjacent shells. It encodes the solutions with first denominator  $a$  by the set

$$\mathcal{W}_a = \{(m, n) \in \mathbb{Z}_{>0}^2 : m \mid pa, n \mid pa, \gcd(m, n) = 1, R_a \mid m + n\},$$

which is in bijection with the ordered solutions  $(a, y, z)$  through  $y = pak/n$ ,  $z = pak/m$ ,  $k = (m + n)/R_a$  (supplement §1, first lemma; this is Theorem 3.1(b) in other coordinates). A *shear edge* is a step  $(a, m, n) \mapsto (a + 1, L(m, n))$  or  $(a, m, n) \mapsto (a + 1, J(m, n))$  with  $(m, n) \in \mathcal{W}_a$  and the image in  $\mathcal{W}_{a+1}$ . The supplement proves, for  $p = 12h + 1$  and shells  $3h + 1 \leq a \leq 9h$ , that no two shear edges compose (Theorem “No two consecutive positive Euclidean shell steps”, publication.tex, line 933). Its proof uses the hypothesis on  $p$  at one step only, and there only through  $p \equiv 1 \pmod{3}$ .

**Proposition 2.1** (the no-two-shears theorem, exact scope). *Let  $p$  be a prime, and consider all shells  $a$  with  $p/4 < a < p$ .*

- If  $p \not\equiv 2 \pmod{3}$ , no two shear edges compose: there is no path  $(a, m, n) \mapsto (a + 1, m', n') \mapsto (a + 2, m'', n'')$  of shear edges.*
- For  $p \equiv 2 \pmod{3}$  such paths exist. At  $p = 131$  one is  $(33, 1, 33) \mapsto (34, 1, 34) \mapsto (35, 1, 35)$ , with  $R_a = 1, 5, 9$ . At  $p = 1613$ , where  $p \equiv 1 \pmod{4}$ , one is  $(404, 1, 404) \mapsto (405, 1, 405) \mapsto (406, 1, 406)$ , with  $R_a = 3, 7, 11$ .*

Status. Generalised here, with the workbench’s proof. The supplement states (a) for  $p \equiv 1 \pmod{12}$  and  $3h + 1 \leq a \leq 9h$  only. Checked on the full graph of every prime below 2000 (script esr\_twoshears.py: no path for  $p \equiv 1 \pmod{3}$ ; 14 paths for  $p \equiv 2 \pmod{3}$ ), and by a complete search up to  $p \leq 10^7$  that uses the shape of an edge proved below: it finds 19,558 paths of two  $L$ -edges, all at primes  $p \equiv 2 \pmod{3}$ , and as many of two  $J$ -edges.

*Proof.* The shape of one edge (the supplement’s classification, whose proof needs only  $a + 1 < p$ ). Let  $(a, m, n) \mapsto (a + 1, m + n, n)$  be an  $L$ -edge. Then  $n$  divides  $pa$  and  $p(a + 1)$ , whose greatest common divisor is  $p$ . If  $n = p$ , then  $p \nmid m$  by coprimality, and  $m + p$  divides  $p(a + 1)$  and is prime to  $p$ , so  $m + p \mid a + 1$ ; but  $m + p > p > a + 1$ . So  $n = 1$ . If  $p \mid m$ , then  $m + 1$  is prime to  $p$  and exceeds  $a + 1$ , and yet divides  $p(a + 1)$ , which is impossible; so  $p \nmid m$  and  $m \mid a$ . Then  $m + 1 \leq a + 1 < p$ , so  $m + 1 \mid a + 1$ . Finally  $R_a \mid m + 1$  and  $R_{a+1} = R_a + 4 \mid m + 2$ . For a  $J$ -edge, exchange  $m$  and  $n$ ; the exchange preserves every  $\mathcal{W}_a$ .

(a) An  $L$ -edge ends at  $(a + 1, m + 1, 1)$ , whose first coordinate is at least 2; a  $J$ -edge starts at a pair whose first coordinate is 1. So an  $L$ -edge is never followed by a  $J$ -edge, and by the exchange a  $J$ -edge is never followed by an  $L$ -edge. Suppose two  $L$ -edges pass through  $(a + i, m + i, 1)$  for  $i = 0, 1, 2$ . By the shape of each edge,  $m + i \mid a + i$  and  $R_a + 4i \mid m + i + 1$  for  $i = 0, 1, 2$ . Put  $K = a - m$ . Then  $m + i \mid K$  for each  $i$ , and one of  $m, m + 1, m + 2$  is divisible by 3, so  $3 \mid K$ . Also  $R_a + 4i$  divides  $4(m + i + 1) - (R_a + 4i) = 4m + 4 - R_a =: C$ , and one of  $R_a, R_a + 4, R_a + 8$  is divisible by 3, so  $3 \mid C$ . But  $C = 4m + 4 - 4a + p = p + 4 - 4K \equiv p + 1 \pmod{3}$ . Hence  $p \equiv 2 \pmod{3}$ . Two  $J$ -edges are excluded by the exchange.

(b) At  $p = 131$ :  $R_{33} = 1$  divides  $1 + 33$ ;  $R_{34} = 5$  divides  $35$ ;  $R_{35} = 9$  divides  $36$ ; each coordinate divides  $pa$ , and each pair is coprime. At  $p = 1613$  (prime):  $3 \mid 405$ ,  $7 \mid 406 = 7 \cdot 58$  and  $11 \mid 407 = 11 \cdot 37$ . Each step is  $J(1, n) = (1, n + 1)$ .  $\square$

So the hypothesis  $p \equiv 1 \pmod{12}$  of the supplement can be weakened to  $p \not\equiv 2 \pmod{3}$ , and not further. The primes  $p \equiv 2 \pmod{3}$  are solvable in any case (Section 3); the proposition concerns the transport graph, not solvability.

## 2.4 The 124-page ES/Fable reader

It encodes a solution of  $4/p = 1/x + 1/y + 1/z$  as the binary quartic

$$H_{\text{ES}}(U, V) = -S^{-1}(U - pV)(U - xV)(U - yV)(U - zV), \quad S = p + x + y + z,$$

with normalised coefficients  $u_0 = -1/S$ ,  $u_2 = -(p(x + y + z) + xy + yz + zx)/S$ ,  $u_3 = 5xyz/S$  and  $u_4 = -pxyz/S$ . It then transports the quartic to the Fable cover and to a weighted conductor of the zeta programme.

**Proposition 2.2.** On solutions,  $p = -5u_4/u_3$  and

$$625u_0u_4^3 - 125u_3u_4^2 + 25u_2u_3^2u_4 - 4u_3^4 = 0.$$

Status. Proved here; the README of the package states both, and the relation was also checked in 21\_ (Proposition 21.6).

*Proof.* Write  $\sigma_1 = x + y + z$ ,  $\sigma_2 = xy + yz + zx$  and  $P = xyz$ . The equation is  $4P = p\sigma_2$ , so the third elementary symmetric function of  $p, x, y, z$  is  $p\sigma_2 + P = 5P$ ; this gives  $u_3$ , and  $p = -5u_4/u_3$  follows. Substituting and multiplying by  $S^4/625$ , the left side becomes  $P^3(p^3 - p^2S + p(p\sigma_1 + \sigma_2) - 4P) = P^3(p\sigma_2 - 4P) = 0$ .  $\square$

The relation is thus the equation itself, written in these coordinates. The Fable and conductor maps built on it are located; their comparison with the zeta programme is in the zeta audit (47\_ overlap OV-13).

## 2.5 The continuation of 17–23 September

The README’s “Start reading” list has 28 items with proof packages in research/incoming/. The results bench R01–R10 and the integration arguments X1–X7 accompany them.

- The first-pass audit 43a\_ lists them all.
- Section 5 states the eleven items audited there.
- Section 6 gives their negative results.

### 3 The core arithmetic, with proofs

Throughout,  $p$  is an odd prime and a *solution* is a triple of positive integers with  $4/p = 1/x + 1/y + 1/z$ . The workbench mostly works with  $p = 12h + 1$  ( $h \geq 1$ ). The results of this section, except Proposition 3.6, were first stated in the reader of four workbenches [9], whose Erdős–Straus section was refereed three times (Appendix B); the audit note is 43\_ and the first-pass report 43a\_.

#### 3.1 The shell criterion

For an integer  $a > p/4$  not divisible by  $p$  put  $R_a = 4a - p$ ,  $S_a = pa$  and

$$E_a = \{u : u \mid a^2, R_a \mid 4u + 1\}, \quad M_a = \{u : u \mid a^2, R_a \mid u + a\}.$$

A shell  $a$  is *occupied* if  $E_a \cup M_a \neq \emptyset$ .

**Theorem 3.1** (shell criterion). (a) *If  $x \leq y \leq z$  is a solution, then  $p/4 < x \leq 3p/4$ . More precisely,  $x < p/2$  and  $x < y$ , unless  $p \equiv 3 \pmod{4}$  and  $(x, y, z) = (\frac{p+1}{2}, \frac{p+1}{2}, \frac{p(p+1)}{4})$ . For  $p = 12h + 1$  this gives  $3h + 1 \leq x \leq 6h$ .*

(b) *For each integer  $a > p/4$  not divisible by  $p$  (in particular each  $a$  with  $p/4 < a < p$ ), the ordered pairs  $(y, z)$  of positive integers with  $1/a + 1/y + 1/z = 4/p$  number exactly  $2|E_a| + |M_a|$ . A pair with  $y = z$  occurs if and only if  $R_a = 1$  (possible only for  $p \equiv 3 \pmod{4}$  and  $a = (p + 1)/4$ ); otherwise  $|M_a|$  is even and the unordered pairs number  $|E_a| + \frac{1}{2}|M_a|$ . For  $p \equiv 1 \pmod{4}$  this is always the case.*

(c) *Hence the equation is solvable at  $p$  if and only if some shell  $a$  with  $p/4 < a < p/2$  is occupied.*

Status. Proved here, and generalised: the workbench’s ES-01/ES-02 (supplement §1; archive §9, “Exact divisor-shell bijection”, p. 38), audited, state it for  $p = 12h + 1$  and  $a \in [3h + 1, 9h]$ ; the proof below gives every odd prime, every  $a > p/4$  with  $p \nmid a$ , and the halved range  $x < p/2$  ( $a \in [3h + 1, 6h]$  for  $p = 12h + 1$ ). The halved range at  $p \equiv 1 \pmod{12}$ , with  $x$  unique, is already a reduction step of the workbench’s item 22, checked by reading in the first pass (43a\_ §2.22), and items 6 and 7 work in it; the workbench’s later packages attribute these coordinates to Elsholtz and Tao [2], §2. Checked against a brute-force count of the ordered pairs on all 4,726 pairs  $(p, a)$  with  $p < 260$  an odd prime and  $p/4 < a < p$  (a pair with  $y = z$  exactly in the 29 cases with  $R_a = 1$ ) and on all 4,303 pairs with  $p < 120$ ,  $p/4 < a \leq 3p$  and  $p \nmid a$ ; on all 4,398 shells of the primes  $p \equiv 1 \pmod{12}$  below 700, and in the first pass on all 8,124 shells below 1000; (a) against all 2,259 solutions at the odd primes below 400; every prime  $p \equiv 1 \pmod{12}$  below 30,000 has an occupied shell.

*Proof.* (a) The other two terms are positive, so  $1/x < 4/p$ ; and  $x$  is the least denominator, so  $4/p \leq 3/x$ . Hence  $p/4 < x \leq 3p/4$ . Suppose  $x > p/2$ , so  $x \geq (p + 1)/2$  as  $p$  is odd, and  $x < p$ . Then  $2/y \geq 1/y + 1/z = 4/p - 1/x > 2/p$ , so  $y < p$ . From  $4xyz = p(xy + yz + zx)$  the prime  $p$  divides  $xyz$ , and as  $0 < x \leq y < p$  it divides  $z = pz'$ ; then  $xy(4z' - 1) = pz'(x + y)$  with  $p \nmid xy$ , so  $4z' - 1 = pk$  with  $k \geq 1$ . Put  $N = pk + 1 = 4z'$ ; then  $1/x + 1/y = 4k/N$ , that is,  $(4kx - N)(4ky - N) = N^2$  with both factors positive. But  $4kx - N \geq 2k(p + 1) - N = N + 2(k - 1)$ , and likewise for  $y$ ; so  $k = 1$  and  $x = y = N/2 = (p + 1)/2$ , with  $z = p(p + 1)/4$ , which needs  $p \equiv 3 \pmod{4}$ . Finally, if  $x < p/2$  and  $y = x$ , then  $1/x + 1/y > 4/p$ , which is impossible. For  $p = 12h + 1$  the integers in  $(p/4, p/2)$  are  $3h + 1, \dots, 6h$ .

(b) Write  $R = R_a$ ,  $S = S_a$ . Then  $R \geq 1$  is odd, because  $p$  is. Since  $p \nmid a$ ,  $\gcd(a, R) = \gcd(a, p) = 1$  and  $\gcd(p, R) = \gcd(p, 4a) = 1$ , so  $S$  is a unit modulo  $R$ . The equation  $1/y + 1/z = R/S$  is equivalent to  $(Ry - S)(Rz - S) = S^2$ , and both factors are positive because  $1/y$  and  $1/z$  are smaller than  $R/S$ . So the ordered pairs correspond bijectively to ordered factorisations  $S^2 = dd'$  into positive divisors with  $d \equiv d' \equiv -S \pmod{R}$ , through  $y = (d + S)/R$ ,  $z = (d' + S)/R$ . If  $d \mid S^2$  and  $d \equiv -S \pmod{R}$ , then  $d$  is a unit and  $d' = S^2/d \equiv S^2(-S)^{-1} = -S$ ; so the pairs are counted by  $\#\{d \mid S^2 : d \equiv -S \pmod{R}\}$ . Since  $\gcd(p, a) = 1$ , each divisor of  $S^2 = p^2 a^2$  is uniquely  $d = p^i v$  with  $i \in \{0, 1, 2\}$  and  $v \mid a^2$ . Using  $4a \equiv p \pmod{R}$ : for  $i = 2$ ,  $p^2 v \equiv -pa \iff pv \equiv -a \iff 4pv \equiv -p \iff R \mid 4v + 1$ , so these  $v$  form  $E_a$ ; for  $i = 1$ ,  $p v \equiv -pa \iff R \mid v + a$ , so these  $v$  form  $M_a$ ; and  $d \mapsto S^2/d$  maps the layer  $i = 2$  bijectively onto the layer  $i = 0$  while preserving the congruence. The total is  $2|E_a| + |M_a|$ . The swap  $(y, z) \mapsto (z, y)$  is  $d \mapsto S^2/d$ ; it maps each layer  $i$  to the layer  $2 - i$ , so a fixed point lies in the layer  $i = 1$ , where it is  $v \mapsto a^2/v$  with only possible fixed point  $v = a$ . That needs  $R \mid 2a$ , i.e.  $R \mid 2$  (as  $\gcd(R, a) = 1$ ), i.e.  $R = 1$ . If  $R \neq 1$  the swap has no fixed point,  $|M_a|$  is even,  $y \neq z$ , and the unordered pairs number  $|E_a| + \frac{1}{2}|M_a|$ . For  $p \equiv 1 \pmod{4}$ ,  $R \equiv -p \equiv 3 \pmod{4}$ , so  $R \neq 1$ .

(c) For  $p \equiv 1 \pmod{4}$  it follows from (a) and (b). For  $p \equiv 3 \pmod{4}$  both sides hold: the shell  $a = (p + 1)/4$  lies in  $(p/4, p/2)$  and has  $R_a = 1$ , so  $u = a$  lies in  $M_a$ .  $\square$

The workbench also gives the exact bookkeeping around this criterion for  $p = 12h + 1$ : an explicit bijection between a tagged set and the ordered solutions, its inverse read off from the  $p$ -adic valuation of  $Ry - S$ , and the global statement that a counterexample prime would satisfy the finite, computable condition  $E_a = M_a = \emptyset$  for every  $a$  (43a\_§1.2; re-derived there).

### 3.2 The first two shells

**Proposition 3.2** (the residual-3 shell). *Let  $a = 3h + 1 = (p + 3)/4$ , so  $R_a = 3$ . Let  $P_1 = \prod (2v_\ell(a) + 1)$  over the primes  $\ell \equiv 1 \pmod{3}$  dividing  $a$ , and  $P_2$  the same product over the primes  $\ell \equiv 2 \pmod{3}$ . Then  $E_a = M_a = \{u \mid a^2 : u \equiv 2 \pmod{3}\}$  and  $|E_a| = |M_a| = P_1(P_2 - 1)/2$ . So a solution with least denominator  $a$  exists if and only if  $a$  has a prime factor  $\equiv 2 \pmod{3}$ , and there are then exactly  $3P_1(P_2 - 1)/4$  unordered solutions with least denominator  $a$ .*

Status. Proved here; the workbench’s residual-3 sieve (supplement §1, `first_two_shell_sieve.tex`; archive §10, “Exact  $R = 3$  factor criterion”), audited. Checked on all 4,472 primes  $p \equiv 1 \pmod{12}$  below  $2 \cdot 10^5$ , and in the first pass on all 19,564 below  $10^6$ .

*Proof.* Since  $a \equiv 1 \pmod{3}$ ,  $3 \nmid a$ . Modulo 3 both  $4u + 1$  and  $u + a$  are  $\equiv u + 1$ , so both conditions say  $u \equiv 2 \pmod{3}$ . A divisor  $u = \prod \ell^{e_\ell}$  of  $a^2$  has  $u \equiv (-1)^{\sum e_\ell} \pmod{3}$ , the sum over the primes  $\ell \equiv 2 \pmod{3}$ . Their exponent vectors  $(0 \leq e_\ell \leq 2v_\ell(a))$  number  $P_2$ , and the difference between the numbers with even and with odd sum is  $\prod_\ell \sum_{e=0}^{2v_\ell} (-1)^e = 1$ ; so  $(P_2 - 1)/2$  have odd sum, and the other primes contribute the factor  $P_1$ . By Theorem 3.1(b) the unordered solutions containing  $a$  number  $\frac{3}{2}P_1(P_2 - 1)/2$ , and  $a = 3h + 1$  is the least denominator of each of them because it is the smallest possible one. Finally  $P_2 > 1$  exactly when  $a$  has a prime factor  $\equiv 2 \pmod{3}$ . (The count is an integer:  $a \equiv 1 \pmod{3}$  makes the total exponent of those primes even, so  $P_2 \equiv 1 \pmod{4}$ .)  $\square$

**Proposition 3.3** (the residual-7 shell). *Let  $a = 3h + 2 = (p + 7)/4$ , so  $R_a = 7$  and  $7 \nmid a$ . Let  $n_3$  be the total exponent in  $a$  of the primes  $\equiv 3 \pmod{7}$  and  $n_{24}$  that of the primes  $\equiv 2$  or  $4 \pmod{7}$ . The shell is occupied if and only if  $a$  has a prime factor  $\equiv 5$  or  $6 \pmod{7}$ , or  $n_3 \geq 3$ , or  $n_3 \geq 1$  and  $n_{24} \geq 1$ .*

Status. Proved here; the workbench’s residual-7 sieve, audited, together with its count as coefficients of  $\prod_\ell \sum_{e=0}^{2v_\ell(a)} T^{\lambda(\ell)e}$  in  $\mathbb{Z}[T]/(T^6 - 1)$ . Checked on all 4,472 primes  $p \equiv 1 \pmod{12}$  below  $2 \cdot 10^5$  (and below  $10^6$  in the first pass).

*Proof.* The exterior condition  $7 \mid 4u + 1$  is  $u \equiv -4^{-1} \equiv 5 \equiv 3^5$ , and the middle condition is  $u/a \equiv -1 \equiv 3^3 \pmod{7}$ ; in base 3 the discrete logarithms of  $1, \dots, 6$  are  $0, 2, 1, 4, 5, 3$ . *Sufficiency.* A prime  $\ell \equiv 5$  dividing  $a$  gives  $u = \ell$ ; a prime  $\ell \equiv 6$  gives  $u = a\ell$ . If  $n_3 \geq 3$ , a product of primes  $\equiv 3$  dividing  $a^2$  with total exponent 5 gives  $u \equiv 3^5$ . If primes  $t \equiv 3$  and  $\ell \in \{2, 4\} \pmod{7}$  divide  $a$ , then  $u = alt$  (when  $\ell \equiv 2 \equiv 3^2$ ) or  $u = a\ell/t$  (when  $\ell \equiv 4 \equiv 3^4$ ) divides  $a^2$  and has  $u/a \equiv 3^3$ . *Necessity.* If none of the conditions holds, every prime factor of  $a$  is  $\equiv 1, 2, 3$  or  $4$ , and either  $n_3 = 0$  or  $(1 \leq n_3 \leq 2 \text{ and } n_{24} = 0)$ . If  $n_3 = 0$ ,  $a$  and all divisors of  $a^2$  lie in the subgroup  $\{1, 2, 4\}$  of squares, which contains neither 5 nor  $-1$ . Otherwise  $a \equiv 3^{n_3}$  and a divisor of  $a^2$  is  $\equiv 3^e$  with  $0 \leq e \leq 2n_3 \leq 4$ ; the exterior target needs  $e \equiv 5 \pmod{6}$  and the middle one  $e - n_3 \equiv 3 \pmod{6}$  with  $|e - n_3| \leq 2$ , both impossible.  $\square$

The shells with  $R = 3$  and  $R = 7$  exist exactly for the primes  $p \equiv 1 \pmod{4}$ , with  $a = (p+3)/4$  and  $a = (p+7)/4$  (Generalised here; checked on all 1,611 such primes below  $3 \cdot 10^4$ ). Proposition 3.3 holds for all of them, since its proof never uses  $p \equiv 1 \pmod{3}$ . Proposition 3.2 holds as stated for  $p \equiv 1 \pmod{12}$  only. For  $p \equiv 5 \pmod{12}$ ,  $a = (p+3)/4 \equiv 2 \pmod{3}$ , so modulo 3 the middle condition reads  $u \equiv 1$ ; then  $E_a = \{u \mid a^2 : u \equiv 2\}$  and  $M_a = \{u \mid a^2 : u \equiv 1\}$ , with  $|E_a| = P_1(P_2 - 1)/2$  and  $|M_a| = P_1(P_2 + 1)/2$  by the parity count of the proof above, and there are exactly  $|E_a| + \frac{1}{2}|M_a| = P_1(3P_2 - 1)/4$  unordered solutions with least denominator  $a$  (an integer: the total exponent in  $a$  of the primes  $\equiv 2 \pmod{3}$  is odd, so  $P_2 \equiv 3 \pmod{4}$ ). That shell is always occupied ( $1 \in M_a$ ), and its occupancy criterion holds trivially, since  $a \equiv 2 \pmod{3}$  has a prime factor  $\equiv 2 \pmod{3}$ ; those primes are solvable anyway, since  $p \equiv 2 \pmod{3}$ .

**Proposition 3.4** (the two-shell survivors). *Let  $p \equiv 1 \pmod{12}$  be prime. Both shells  $a = (p+3)/4$  and  $a = (p+7)/4$  are unoccupied if and only if  $p \equiv 1 \pmod{24}$ , every prime factor of  $(p+3)/4$  is  $\equiv 1 \pmod{3}$ , and every prime factor of  $(p+7)/4$  is  $\equiv 1, 2$  or  $4 \pmod{7}$ . Every such  $p$  is a square modulo 7, so  $p \equiv 1, 25$  or  $121 \pmod{168}$ .*

Status. Proved here; the first pass stated the congruence mod 24 as an observation on its computation. Checked on all primes  $p \equiv 1 \pmod{12}$  below  $2 \cdot 10^5$ .

*Proof.* If  $p \equiv 13 \pmod{24}$ , then  $(p+3)/4$  is even and  $2 \equiv 2 \pmod{3}$ , so the first shell is occupied (Proposition 3.2). If  $p \equiv 1 \pmod{24}$ , the first shell is unoccupied exactly when every prime factor of  $(p+3)/4$  is  $\equiv 1 \pmod{3}$ ; and  $(p+7)/4$  is even, so  $n_{24} \geq 1$  and Proposition 3.3 says that the second shell is occupied exactly when  $(p+7)/4$  has a prime factor  $\equiv 3, 5$  or  $6 \pmod{7}$ , a non-residue. If it has none,  $(p+7)/4$  is a square modulo 7, and so is  $p \equiv 4 \cdot \frac{p+7}{4}$ . Among the classes  $\equiv 1 \pmod{24}$  modulo 168, the squares modulo 7 are 1, 25 and 121.  $\square$

Of the primes  $p \equiv 1 \pmod{12}$  below  $10^6$ , exactly 989 have both shells unoccupied, the first being 1129, 1201, 2521 (first pass).

### 3.3 Which primes remain

The workbench proves solvability for  $p \equiv 2 \pmod{3}$ ,  $p \equiv 3 \pmod{4}$  and  $p \equiv 13 \pmod{24}$  (scope identities that it does not advertise as new) by the families  $(p, \frac{p+1}{3}, \frac{p(p+1)}{3})$ ,  $(\frac{p+1}{4}, \frac{p(p+1)}{2}, \frac{p(p+1)}{2})$  and  $(a, \frac{pa}{2}, pa)$  with  $a = \frac{p+3}{4}$  (even when  $p \equiv 13 \pmod{24}$ ); so its own reduction leaves the primes  $p \equiv 1 \pmod{24}$ . Its later packages restrict to the *hard* primes  $p \equiv 1, 121, 169, 289, 361, 529 \pmod{840}$ , which are exactly the unit squares modulo 840 (a group of order  $1 \cdot 1 \cdot 2 \cdot 3 = 6$ ). The reduction to them is classical (Obláth and Mordell; see Elsholtz–Tao [2] and Mordell [5]). The 22 September packages use it without proof; the archive proves it (§10, “Non-square modulo 840 theorem”, p. 95). Five identities prove it.

**Proposition 3.5** (the reduction modulo 840). *Let  $p \equiv 1 \pmod{24}$  be prime, and let  $(A, B, R)$  be  $(1, 2, 7)$ ,  $(2, 1, 7)$ ,  $(1, 1, 7)$ ,  $(1, 2, 15)$  or  $(2, 1, 15)$  according as  $p \equiv 3, 5, 6 \pmod{7}$  or  $p \equiv 2, 3 \pmod{5}$ . Then  $D = (p + R)/(4AB)$  and  $C = (A + pB)/R$  are positive integers and*

$$\frac{4}{p} = \frac{1}{ABD} + \frac{1}{ACD} + \frac{1}{pBCD}.$$

*So the equation is solvable at every prime  $p \equiv 1 \pmod{24}$  that is not a square modulo 840.*

Status. Proved here (the reduction is classical); the first pass covered the 18 non-square unit classes  $\equiv 1 \pmod{24}$  by the same kind of families. Checked on all primes  $p \equiv 1 \pmod{24}$  below  $3 \cdot 10^5$ .

*Proof.* With  $RC = A + pB$  and  $4ABD = p + R$ :  $\frac{1}{ABD} + \frac{1}{ACD} + \frac{1}{pBCD} = \frac{pC + pB + A}{pABCD} = \frac{C(p+R)}{pABCD} = \frac{p+R}{pABD} = \frac{4}{p}$ . Since  $p \equiv 1 \pmod{8}$ , 8 divides  $p + 7$  and  $p + 15$ , so  $4AB \mid p + R$ ;  $R \mid A + pB$  is the stated congruence modulo 7 or 5, together with  $A + pB \equiv A + B \equiv 0 \pmod{3}$  when  $R = 15$ . The primes  $p \equiv 1 \pmod{24}$  left over have  $p \equiv 1, 2, 4 \pmod{7}$  and  $p \equiv \pm 1 \pmod{5}$ , which are exactly the unit squares modulo 840.  $\square$

### 3.4 The Jacobi-symbol barrier

**Proposition 3.6** (the Jacobi-symbol barrier). *Let  $p$  be an odd prime,  $R \equiv 3 \pmod{4}$ ,  $a = (p + R)/4$  an integer, and  $\gcd(R, pa) = 1$ . If the Jacobi symbol  $(q/R)$  equals 1 for every prime  $q \mid a$ , then the shell  $a$  is unoccupied. The converse is false: at  $p = 5$ ,  $R = 7$ ,  $a = 3$  the only prime factor of  $a$  is a non-residue modulo 7, and yet the shell is unoccupied.*

Status. Proved here; archive §10, “Jacobi-symbol barrier” and “The Jacobi barrier has no converse” (p. 96), whose proofs are correct as read.

*Proof.* By the proof of Theorem 3.1(b), the shell is occupied exactly when some divisor  $d$  of  $S^2 = p^2a^2$  satisfies  $d \equiv -pa \pmod{R}$ . Let  $\chi = (\cdot/R)$  be the Jacobi symbol, a character on the units modulo  $R$ . The hypothesis gives  $\chi(a) = 1$ , and  $p \equiv 4a \pmod{R}$  gives  $\chi(p) = \chi(4)\chi(a) = 1$ . So  $\chi(d) = 1$  for every divisor  $d$  of  $p^2a^2$ , while  $\chi(-pa) = \chi(-1) = (-1)^{(R-1)/2} = -1$  because  $R \equiv 3 \pmod{4}$ . So no divisor lies in the target class. For the converse: the squares modulo 7 are 1, 2, 4, so  $(3/7) = -1$ ; the divisors of  $15^2 = 225$  are  $\equiv 1, 2, 3, 4, 5 \pmod{7}$ , and the target  $-15 \equiv 6$  is not among them.  $\square$

So at a prime  $p \equiv 1 \pmod{4}$ , a shell with residual  $R$  can be occupied only if  $a = (p + R)/4$  has a prime factor that is a non-residue modulo  $R$  in the Jacobi sense. The argument is a quadratic-character argument: every divisor inherits the residue condition. This is the kind of mechanism behind the obstruction to polynomial identities recalled in Section 1. The barrier explains why the workbench’s occupancy criteria, such as Propositions 3.2 and 3.3, are phrased through non-residue prime factors of  $a$ .

## 4 Least classes, record primes, explicit families, and the carrier census

### 4.1 The least-class function and its record primes

The archive (§17.4 and §19, pp. 131 and 148–157) puts, for  $i \geq 1$ , following Ionascu and Wilson [3],

$$C_i = \left\{ n \geq 2 : \frac{4}{n} = \frac{1}{x} + \frac{1}{y} + \frac{1}{z} \text{ with } x \leq \frac{n + 4i - 1}{4} \right\}, \quad h(n) = \min\{i : n \in C_i\}.$$

An odd prime  $p$  is a *strict record* if  $h(q) < h(p)$  for every prime  $q < p$ .

**Lemma 4.1.** *Let  $p \equiv 1 \pmod{4}$ . The least denominator occurring in any solution at  $p$  is the least occupied shell  $a$ . With  $R = 4a - p$  one has  $h(p) = (R + 1)/4$ . For  $p \equiv 3 \pmod{4}$ ,  $h(p) = 1$ , and  $h(2) = 1$ .*

Status. Proved here.

*Proof.* If  $a$  is occupied, some solution has  $a$  as a denominator, so the least denominator of that solution is at most  $a$ . Conversely, the least denominator  $x$  of a solution is itself an occupied shell: it exceeds  $p/4$  (Theorem 3.1(a)) and is completed by the other two denominators. So the least occupied shell equals the least denominator over all solutions. A solution lies in  $C_i$  exactly when  $4x - p \leq 4i - 1$ . For  $p \equiv 1 \pmod{4}$  the residual  $4x - p$  is  $\equiv 3 \pmod{4}$ , so the least  $i$  is  $(R + 1)/4$ . For  $p \equiv 3 \pmod{4}$  the shell  $(p + 1)/4$  is occupied (Theorem 3.1(c)). For  $p = 2$ , take  $4/2 = 1/1 + 1/2 + 1/2$  with  $x = 1 \leq 5/4$ .  $\square$

**Proposition 4.2** (the record primes up to 8,803,369). *The strict records  $p \leq 8,803,369$ , with  $h(p)$  and  $R = 4h(p) - 1$ , are exactly*

$(2, 1, 3)$ ,  $(73, 2, 7)$ ,  $(1129, 3, 11)$ ,  $(1201, 6, 23)$ ,  $(21169, 8, 31)$ ,  $(118801, 15, 59)$ ,  $(8803369, 27, 107)$ .

Moreover  $h(806521) = h(118801) = 15$ , so 806,521 is not a strict record.

Status. Reproduced here: every prime up to 8,803,369 was computed by Lemma 4.1 and Theorem 3.1, using the residues of the divisors of  $a^2$  modulo  $R$  (script `esr_records.py`, under three seconds). This agrees with the archive’s table (§19.2, p. 152) and with its value  $1201 \in C_6 \setminus C_5$  (§17.4, p. 132), which the archive records without claiming it as new. At  $p_* = 8,803,369$  the least occupied shell has  $R = 107$ , the shell of archive §3.

For the five records whose residual  $R$  is a safe prime ( $R = 2m + 1$  with  $m$  prime), and whose shell  $a = (p + R)/4$  has exactly one non-residue prime factor modulo  $R$ , to exponent one, the archive defines a *deficit* set  $D \subset \mathbb{Z}/m$  from the discrete logarithms of the prime factors of  $a$  (§19.1). It states the *strict-record deficit-box conjecture* (§19.2): at such a record with non-empty deficit,  $D$  is a difference-proper generalised arithmetic progression of rank at most two, and two induced “odd sheets” cover  $\mathbb{Z}/m$ . The deficits of the first three rows are empty. The two non-empty ones are proved to be a rank-two rectangle  $\{11, 14, 15, 18\} \subset \mathbb{Z}/29$  at  $(118801, 59)$  and the progression  $\{23 + 42j : 0 \leq j \leq 9\} \subset \mathbb{Z}/53$  at  $(8803369, 107)$ . The boundary example  $p = 806,521$  satisfies every hypothesis except strict recordness, and both conclusions fail there. So the hypothesis cannot be dropped.

Status. The conjecture is open, as the archive says. Its finite evidence is reproduced here from the definitions (script `esr_deficits.py`, with the primitive root  $g = 2$  that the archive uses for  $R = 59$  and  $R = 107$ ):

- the deficits of the rows  $p = 73, 1129$  and  $1201$  are empty, and the row  $p = 21169$  is outside the domain because  $m = 15$  is composite;
- at  $p = 118,801$ :  $a = 29715 = 3 \cdot 5 \cdot 7 \cdot 283$ ,  $D = \{11, 14, 15, 18\}$ ,  $D - D = \{3r + 4s : |r|, |s| \leq 1\}$  with nine elements, offsets  $u = 11$  and  $v = 17$ , and the two odd sheets cover  $\mathbb{Z}/29$ ;
- at  $p = 8,803,369$ :  $a = 2,200,869 = 3^2 \cdot 11^2 \cdot 43 \cdot 47$ ,  $D = \{23 + 42j : 0 \leq j \leq 9\}$ ,  $D - D = \{42k : |k| \leq 9\}$  with nineteen elements, offsets 29 and 23, and the sheets cover  $\mathbb{Z}/53$ ; also the corollary of archive §16, that the middle target  $-1$  is reached by exactly two exponent vectors and the exterior target not at all;
- at  $p = 806,521$ :  $a = 201,645 = 3^2 \cdot 5 \cdot 4481$ , a deficit of 14 points with  $D - D = \mathbb{Z}/29$ , and the odd sheets miss exactly  $\{1, 4, 8, 12, 16, 20, 24, 27\}$ .

Another primitive root multiplies  $D$  by a unit of  $\mathbb{Z}/m$  (archive §19.1). The script confirms that the sizes of  $D$  and  $D - D$ , and the number of missed points, do not change. The failure of conclusion (i) at 806,521 follows from  $D - D = \mathbb{Z}/29$  by the archive’s counting argument. A difference-proper progression of rank one with 14 terms has exactly 27 differences. A proper one of rank two has side lengths  $\{2, 7\}$ , so a difference-proper one would have  $3 \cdot 13 = 39 > 29$  distinct differences. That argument was read and

is correct. By Theorem 3.1(b), the counts at  $p_* = 8,803,369$  ( $|E_a| = 0$ ,  $|M_a| = 2$ ) mean that exactly one unordered solution has least denominator 2,200,869, namely

$$\frac{4}{8803369} = \frac{1}{2200869} + \frac{1}{181085300330} + \frac{1}{3293760527702370}.$$

The archive’s Lean module covers the  $R = 107$  row only, as the archive states.

## 4.2 Four progressions on which the equation is solved

**Theorem 4.3** (four unconditional families). *For  $k \geq 0$  put  $g_1 = 31k+30$ ,  $g_2 = 31k+15$ ,  $m_3 = 23k+15$ ,  $m_4 = 23k+19$ ,  $w_3 = 1116k+727$  and  $w_4 = 4464k+3683$ . Then  $4/n_i = 1/a_i + 1/b_i + 1/c_i$  with positive integers*

| $i$ | $n_i$          | $a_i$    | $b_i$         | $c_i$           |
|-----|----------------|----------|---------------|-----------------|
| 1   | $248k+209$     | $2g_1$   | $2n_1(k+1)$   | $2n_1g_1(k+1)$  |
| 2   | $248k+89$      | $2g_2$   | $n_2(2k+1)$   | $2n_2g_2(2k+1)$ |
| 3   | $17112k+11137$ | $186m_3$ | $12n_3m_3w_3$ | $124m_3w_3$     |
| 4   | $17112k+14113$ | $186m_4$ | $6n_4m_4w_4$  | $31m_4w_4$      |

*No primality of  $n_i$  is needed.*

Status. Proved here by direct expansion:  $4a_ib_ic_i - n_i(b_ic_i + a_ic_i + a_ib_i)$  is the zero polynomial in  $k$  for each  $i$  (sympy), and the entries are positive for  $k \geq 0$ ; exact fractions agree for  $k \leq 200$  (script `esr_families.py`). Archive §24, p. 423. The archive shows that these triples are specialisations of Rosati’s two 1954 parametrisations and claims no independence from the classical parametrisations (archive abstract, p. 2, and §13.1). That comparison is located, not checked.

*Proof.* All displayed linear factors have positive constant terms, so every entry is a positive integer for  $k \geq 0$ . The identity  $4a_ib_ic_i = n_i(b_ic_i + a_ic_i + a_ib_i)$  is a polynomial identity in  $k$ , verified by expanding both sides. Dividing by  $n_ia_ib_ic_i$  gives the decomposition.  $\square$

Each progression meets all six hard classes modulo 840. Since  $\gcd(248, 840) = 8$  and  $\gcd(17112, 840) = 24$ , the numbers  $n_1, n_2$  run through every class  $\equiv 1 \pmod{8}$  modulo 840, and  $n_3, n_4$  through every class  $\equiv 1 \pmod{24}$ . Every hard class is  $\equiv 1 \pmod{24}$ , being the square of a unit. Each  $n_i(0)$  is prime to the step of its progression, so by Dirichlet’s theorem every progression contains infinitely many primes in each hard class. Examples are the strict records  $1201 = n_1(4) \equiv 361$  and  $21169 = n_2(85) \equiv 169 \pmod{840}$ . So the families solve the equation at infinitely many primes in every class that the reduction modulo 840 leaves open (script `esr_families.py` lists the least prime for each family and class). This does not conflict with the obstruction to polynomial identities recalled in Section 1. That obstruction is modulo  $q$ , and it applies only when  $r$  is a square modulo  $q$ : here  $11137 \equiv 5 \pmod{23}$ , and 5 is not a square modulo 23. The same holds for the other rows, since  $14113 \equiv 14 \pmod{23}$ ,  $209 \equiv 23 \pmod{31}$  and  $89 \equiv 27 \pmod{31}$  are non-squares.

## 4.3 The carrier census

The largest part of the archive’s arithmetic (§18 and the second half of §24, about 170 pages) is a sequence of *sufficient-carrier censuses*.

- *Rows.* For the hard primes in fixed branches, the archive considers residue data of  $p$  modulo a growing product of moduli. We call each such datum a *row*. The moduli are 11088, 5, 19, 23, 31 and 47, then 71, 191, 83, 167 and 239.

- *Forced rows.* A row is *forced* when an explicit construction shows that, for every prime  $p$  in the branch with that row, one of a finite list of shells is occupied. An example is the implication (2272) on p. 431: for a prime  $p \equiv 25 \pmod{72}$  whose shells  $R = 7$  and  $R = 11$  are empty and whose row lies in the domain but outside the survivor set, some shell with  $R \in \{15, 19, 23, 31, 47\}$  is occupied.
- *Survivors.* The remaining rows are *survivors*. Nothing is concluded for them.

The archive’s front section (pp. 3–5) reports the following chain.

| Step                                  | Domain (rows)          | Forced                 | Survivors              |
|---------------------------------------|------------------------|------------------------|------------------------|
| six shells after the unit lift mod 71 | 4,590,432,000          | 4,212,008,640          | 378,423,360            |
| $R = 191$                             | 71,900,438,400         | 24,762,018,720         | 47,138,419,680         |
| $Q = 83$                              | 3,865,350,413,760      | 934,648,428,742        | 2,930,701,985,018      |
| $R = 167$                             | 486,496,529,512,988    | 115,851,891,806,170    | 370,644,637,706,818    |
| $R = 239$                             | 88,213,423,774,222,684 | 20,524,419,041,579,843 | 67,689,004,732,642,841 |
| refresh of the $R = 167$ coefficient  |                        | 39,025,685,947,842     | 67,649,979,046,694,999 |

Cumulatively 2,757,919,929,517,785,001 of the 2,825,569,908,564,480,000 rows of the final hard domain are forced, a proportion of 0.97606 .... The table is internally consistent (checked here):

- each domain is the previous survivor count times  $q-1$ , for the next modulus  $q = 191, 83, 167, 239$ ;
- the final domain is  $4,590,432,000 \cdot 190 \cdot 82 \cdot 166 \cdot 238$ ;
- forced plus survivors equals the domain in every row;
- the fractions 69643/75900 and 6257/75900 of the first row are exact.

Status. Located. The arithmetic consistency of the reported numbers was checked here; the carrier theorems and the enumerations behind them were not re-run. The archive states the following about its own checks.

- The enumerations are replayed by independent Python programs.
- Bounded Lean modules check the literal census rows, totals, fractions, target residues and candidate orders, but not the affine and gcd reconstructions, carrier membership or the positive-denominator witnesses (pp. 3–5).
- The ranking that selected 239 after 167 was corrected: on the full ten-coordinate object, 59 ranks first and 239 second. The  $R = 239$  step is kept as a valid non-greedy continuation, not as an optimal one (p. 4).

The archive calls these “bounded sufficient-carrier censuses” and states that they are not shell-exhaustion, density, endpoint or optimality theorems (p. 2). A survivor row is only outside the carriers’ images; it is not shown to be shell-empty.

Two remarks help a reader place the census.

- It is a covering argument of the same general kind as Mordell’s identities, refined by factor and gcd data. Each forced row is certified by explicit divisor constructions, not by polynomial identities alone.
- The survivors make up 2.39% of the final domain. Each of the four lifts forced between 23% and 35% of its domain ( $R = 191$ : 34.4%;  $Q = 83$ : 24.2%;  $R = 167$ : 23.8%;  $R = 239$ : 23.3%). By Proposition 3.6, a single shell  $R$  can be occupied at  $p$  only if  $(p + R)/4$  has a prime factor that is a non-residue modulo  $R$ , which is not a condition on residues of  $p$  alone.

Whether the survivors can be removed in this way is the open global obligation named by the archive itself (p. 4).

## 5 The September continuation

The continuation of 17–23 September is organised as the 28-item “Start reading” list of the README, with proof packages in research/incoming/. The workbench describes the results of its 22 September release as written proofs with Python replays, without a Lean build or independent human review (43a\_ §2, citing DAILY\_RESULTS\_20260922).

### 5.1 The audited items

The 28 items of the README’s “Start reading” list (17–23 September) are listed in 43a\_ §2, with their packages. Those audited:

- *Item 6* (audited). With  $L_a = 8\tau(a)^2/\varphi(R_a) - \mathcal{E}_a$ , where  $\mathcal{E}_a$  is the residue-collision energy of the  $2\tau(a)$  divisors of  $pa$  modulo  $R_a$ : for  $p \equiv 1 \pmod{4}$ ,  $p \geq 2^{20}$ ,  $\sum_{a \in (p/4, p/2)} L_a \leq -\frac{1}{24}p \log p$ . So the lower bound that keeps only the principal character, summed over the shells, is negative for large  $p$ : that method cannot prove occupancy.
- *Item 7* (audited). For  $p \equiv 1 \pmod{8}$ , bounds in terms of the least quadratic nonresidue; at the hard primes a middle state  $(a, u)$  on a shell  $p/4 < a < p/2$  has  $a \leq 3(p+3)/11$ , and an exterior state satisfies  $22(p+1) \leq 23(p-2a+1)^2$ . The workbench says this localises solutions and does not prove the region occupied.
- *Item 14* (audited). For  $h' = 4j - 1 > t$  the divisors  $W \mid j^2$  with  $W \equiv t \pmod{h'}$  are canonical, companion or finite templates; the finite ones satisfy the sharp bound  $h' \leq t^2 - 3t + 1$ , with equality exactly in one parametrised family, attained at infinitely many primes  $p \equiv 1 \pmod{840}$ . For every  $t \geq 4$  infinitely many hard primes have an exterior state whose same-grade middle box is empty.
- *Item 21* (audited). For a proper rational trace source the complete set of same-word residual returns is  $\{k \mid R : d \mid k, k \equiv 1 \pmod{4K(u)}\}$ ; a uniform return exists exactly for the nine words  $u \mid 36$ .
- *Item 22* (audited, certificate). For every solution at  $p \equiv 1 \pmod{12}$  an explicit symmetric polynomial  $\mathfrak{N}(p, x, y, z)$  of degree 8 satisfies  $\mathfrak{N} < -\frac{125873811}{262144}p^8$  ( $\approx -480.17p^8$ ), and  $\mathfrak{N} < -6960.19p^8$  when  $p \equiv 1 \pmod{24}$ ; the proof reduces to finite polynomial positivity certificates, recomputed independently. It is a statement about existing solutions, not about their existence.
- *Item 26* (audited in part: the least example and the deduction from it; its prime-square trace theorem at  $p \equiv 1 \pmod{24}$  and its terminality statement read and correct as read, with the identity  $16(4q^3+1) - (4q+1)(16q^2-4q+1) = 15$  checked; the factor-sum, Pell and fixed-tail-sum sections not checked).  $p = 67369$  is the least prime  $p \equiv 1 \pmod{24}$  whose first “trace shell” ( $a = 16849, R = 27$ ) precedes its first full shell ( $a = 16850, R = 31$ ); so no universal map from rational traces to solutions avoids increasing  $a$ .
- *Item 4* (conditional on a computer enumeration of 1,381,117,764 rooted profiles, 5,922,259 of them range-compatible, run by two implementations of the workbench and not re-run here). At every hard prime, from any nonresidue prime, factoring at most 25 numbers  $(p + \sigma q t^2)/4$  with  $t \in \{1, 3, 5, 7, 9\}$  yields at least six distinct nonresidue primes. A direct test of all 108 hard primes below 40,000 agrees.

Items 2 and 3 are method counterexamples, and their certificates are audited: the seven-cycle sink of item 2 at  $p = 2521$  and the closed triangle of item 3; their closure and escape theorems are located. For item 8, the middle identity  $p = RQ - (Q+1)^2/(4A)$  and its two bounds were reproduced on all 1,613 oriented middle states at primes  $p \equiv 1 \pmod{8}$  below 4000, and its atlas is located. The 22-digit prime 7510085481569082811681 of item 3 is proved prime here by a Pocklington chain  $(p-1 = 2^5 \cdot 3 \cdot 5 \cdot q_1, q_1-1 = 2 \cdot 3 \cdot 5 \cdot 351707 \cdot q_2, q_2-1 = 2 \cdot 3 \cdot 17^2 \cdot 5051 \cdot 169307, \text{witnesses below } 12 \text{ at every level, the leaves by trial division})$ . Item 23 is audited in part (its

certificates; the deduction read in part); items 11–13, 15 and 20 use imported material (Section 7); the remaining 17 items, including these five, are located. The results bench R01–R10 and the arguments X1–X7 are checked in 43a\_ §3; one source passage error (X5:  $289^3 \equiv 169$ , not  $-1$ , modulo 840) is recorded by the workbench itself.

## 5.2 The $(3, 4, \infty)$ monodromy covers

A package of the workbench (ES-09) imports three integer matrices  $A_1, A_2, A_\infty$  with  $A_1^3 = A_2^4 = A_1 A_2 A_\infty = I$  from the  $S^6$  manuscript [1] and nothing else from it. For odd levels  $D$ , which is the source’s standing hypothesis, the induced affine action on  $(\mathbb{Z}/D)^3$  is transitive when  $3 \nmid D$  and has two orbits, of sizes  $D^3/9$  and  $8D^3/9$ , when  $3 \mid D$ ; for  $3 \nmid D$  the cover is connected of genus  $(5D^3 - 12D^2 - 17D + 24)/24$ , and for  $3 \mid D$  the two orbits give components of genera  $(5D^3 - 12D^2 - 81D + 216)/216$  and  $(5D^3 - 12D^2 - 9D + 27)/27$ . These statements are false for even  $D$  (at  $D = 2$  the orbits have sizes 2 and 6; found by the referee of the notes), which the ES application never uses, since there  $D$  divides the odd number  $R_a$ . The package also shows that the equation fails at  $p$  if and only if a denominator-labelled cover has no section. That equivalence is correct, and it is Theorem 3.1 again: a section exists exactly when the tail denominator  $R_a / \gcd(R_a, 4u + 1)$  or  $R_a / \gcd(R_a, u + a)$  equals 1.

Status. Audited: the proof read; the relations, the orbits and the genera also recomputed from cycle counts for  $D = 1, 3, \dots, 15$ . The equivalence is a re-encoding of Theorem 3.1; the assessment is this audit’s (the package presents it as a complete failure object).

## 6 Negative results, with exact scope

Each row says which proposed implication fails and what the failure does not show. Every certificate prime in the table is itself solvable.

| Source                           | What fails                                                                        | Certificate                                                             | What it does not show                        |
|----------------------------------|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------|----------------------------------------------|
| bounded transport                | every occupied shell has a solution with unary layer<br>$A \leq 2$                | $p = 37, a = 12$ : only<br>$u = 8$                                      | the global cutoff<br>( $a = 10$ works at 37) |
| first-two-shell sieve            | occupancy of the $R = 7$ shell<br>gives a first-two-layer hit                     | $p = 373, a = 95$                                                       | anything about other<br>shells               |
| item 2                           | every canonical closed<br>component contains a local hit                          | $p = 2521$ : a 7-cycle<br>sink                                          | 2521 is solvable                             |
| item 3                           | strict canonical descent ends at<br>a hit                                         | a closed triangle at a<br>22-digit prime                                | that prime is solvable                       |
| item 6                           | the principal-only bound,<br>summed over shells, is<br>positive for large $p$     | $\sum L_a \leq -\frac{1}{24}p \log p$                                   | anything about the<br>true counts            |
| item 6                           | a nonnegative reweighting of<br>it certifies occupancy                            | $p = 87481$ : all 21,870<br>shells have $L_a \leq 0$                    | only this certificate<br>family fails        |
| item 14                          | same-grade middle repair<br>from exterior states with<br>$\alpha = -4t, t \geq 4$ | $p = 7840561$                                                           | those primes are<br>solvable                 |
| item 21                          | a same-word residual return<br>for words $u \nmid 36$                             | 3,041 of the 3,654<br>sources with $u \nmid 36$ at<br>primes below 2600 | endpoint solutions<br>exist                  |
| item 26                          | a trace-to-solution map never<br>increases $a$                                    | $p = 67369$                                                             | 67369 is solvable                            |
| record shells<br>(archive §19.3) | the deficit-box conclusions<br>without strict recordness                          | $p = 806,521$ : full<br>difference set, eight<br>points missed          | anything about strict<br>records             |

| Source                                      | What fails                                                                                                                   | Certificate                                   | What it does not show                                                                          |
|---------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|------------------------------------------------------------------------------------------------|
| Jacobi barrier<br>(archive §10)             | its converse: a non-residue prime factor of $a$ makes the shell occupied                                                     | $p = 5, R = 7, a = 3$                         | the barrier itself<br>(Proposition 3.6)                                                        |
| no-two-shears<br>theorem<br>(supplement §5) | the theorem at primes $p \equiv 2 \pmod{3}$ , with its definitions extended to all shells<br>$p/4 < a < p$ (Proposition 2.1) | two-edge paths at<br>$p = 131$ and $p = 1613$ | anything at $p \not\equiv 2 \pmod{3}$ , where the theorem holds;<br>anything about solvability |

All of these are statements about methods or certificate families. None is a counterexample to the conjecture, and none bears on whether it is true.

The archive records further negative results of the same kind. They are located, not checked here:

- the withdrawn constructions of §5 (p. 22), each with its point of failure;
- the retracted converse of the Star–Kneser forcing theorem (§15.3, p. 122);
- the failed universal bridge of Dyachenko’s bounded identities (§12.2, p. 103);
- a finite counterexample to one projection step in López Bonilla’s 2026 preprint (§19, provenance, p. 148): in  $C_6$ , the set  $\{0, 1\}$  has trivial stabiliser, but its reduction modulo 2 is all of  $C_2$ . The archive restricts this to that step and does not assess the whole paper.

## 7 Where the workbench meets the author’s other workbenches

The owner notes that the Erdős–Straus and zeta sessions often ran at the same time, and that material crossed between them. The note 47\_ records every crossing that the audit found as a typed edge. Its edge types are:

- $P$ : a proof-level import;
- $D$ : a dependency or citation;
- $I$ : the same identity or object on both sides;
- $A$ : an analogy only.

The table lists the edges that touch this workbench, and adds three that this reader found in the archive (marked “new”).

| Edge                       | What crosses                                                                                                                                                                                                                                                                                                          | Type | Status                                                 |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|--------------------------------------------------------|
| NS $\rightarrow$ ES        | The auxiliary torus endomorphism<br>$J = \begin{pmatrix} 3 & 1 \\ 1 & 5 \end{pmatrix}$ (determinant 14) of the<br>Navier–Stokes manuscript. It enters 21<br>statements of the supplement (§8) as an<br>operator only. The core identity is $\{(u, v) \in$<br>$G^2 : u^3v = 1 = uv^5\} = \{(u, u^{-3}) : u^{14} = 1\}$ | P    | audited (21_ §1); no fluid<br>theorem is used          |
| $S^6 \rightarrow$ ES       | The integral monodromies $A_1, A_2, A_\infty$ of the<br>(3, 4, $\infty$ ) period system, in ES-09                                                                                                                                                                                                                     | P    | audited for odd levels (§5.2)                          |
| ES $\leftrightarrow$ $S^6$ | The Niemeier lattice with root system $A_5^4 D_4$<br>and glue index 72 (ES result R10)                                                                                                                                                                                                                                | I    | ES’s glue code verified; the<br>two codes not compared |

| Edge                                                | What crosses                                                                                                                                                                                                                                                                                                                                                | Type | Status                                                                                                     |
|-----------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|------------------------------------------------------------------------------------------------------------|
| Jacobian map                                        | Alpöge’s map $F : \mathbb{C}^3 \rightarrow \mathbb{C}^3$ with $\det DF = -2$ and the three-point fibre over $(-\frac{1}{4}, 0, 0)$ (archive §§4, 7). The same map is in the Yang–Mills workbench and the zeta reader. ES’s four-dimensional extension $P$ ( $\det DP = -2$ ) crosses to the zeta programme’s conductor                                      | P    | $\det DF$ , $\det DP$ and the fibre verified (zeta reader [8], Lemma 6.6; 21_); the conductor maps located |
| ES $\leftrightarrow$ zeta                           | The split-zero semiring $G(R) = R \sqcup \{\tau\}$ with its Boolean support character (archive §21; ES R01)                                                                                                                                                                                                                                                 | I    | audited (21_ §6)                                                                                           |
| ES $\leftrightarrow$ zeta                           | A finite Weyl phase-space basis                                                                                                                                                                                                                                                                                                                             | I    | ES side verified on $C_2 \times C_3$                                                                       |
| zeta $\rightarrow$ ES                               | NG20–NG21, used by item 20; the SplitZero packages of 13–16 September                                                                                                                                                                                                                                                                                       | D    | located                                                                                                    |
| ES $\rightarrow$ zeta                               | The 488-page reader’s source, vendored into the zeta repository                                                                                                                                                                                                                                                                                             | D    | located                                                                                                    |
| ES $\rightarrow$ zeta                               | The 124-page ES/Fable reader: solutions as binary quartics, then the Fable cover and a weighted conductor (§2.4)                                                                                                                                                                                                                                            | P    | the quartic relation proved here (Proposition 2.2); the rest located                                       |
| ES $\leftrightarrow$ Collatz                        | $3(4n+1)+1 = 4(3n+1)$ , i.e. $\theta T = 4\theta$ (ES R05, diagrams by u/CivQ17)                                                                                                                                                                                                                                                                            | I    | audited                                                                                                    |
| Collatz $\leftrightarrow$ ES $\leftrightarrow$ zeta | The homogenisation $(x, h) \mapsto (ax + bh, h)$ (ES X4)                                                                                                                                                                                                                                                                                                    | I    | ES’s diagram read                                                                                          |
| Busy Beaver (new)                                   | Busy-Beaver extrema quoted by the archive ( $S(4) = 107$ and $\text{Space}(5) = 12289$ ; for the five-state values it cites the formal enumeration of the bbchallenge collaboration) select the shells of archive §§3 and 20; §25 maps extrema to shell carriers. The zeta reader’s detectability argument uses a different machine for a different purpose | A    | located; no mathematical dependence between the two                                                        |
| Connes (new)                                        | Supplement §6 reads Connes’ primitive intersections; archive §24 opens with Tomita–Takesaki transport. The zeta programme uses Connes’ summation map and Meyer’s quotient                                                                                                                                                                                   | A    | located; shared names, no common statement found                                                           |
| Fable (new)                                         | “Fable” names three different objects across the repositories: the three-dimensional Keller map, the $S^6$ construction in older files, and a conductor cover (47a_ §10). Archive §5 withdraws one Fable construction                                                                                                                                       | —    | disambiguate at each use                                                                                   |
| ES $\leftrightarrow$ YM                             | The word “gap” (ES X7), and a proposed success projector                                                                                                                                                                                                                                                                                                    | A    | no mathematical content                                                                                    |

So the mathematical crossings are few and specific: the torus operator, the monodromy matrices, the Keller maps, the split-zero semiring, and two identities. Each is either audited or recorded as located. The remaining connections are shared vocabulary or proposals. The archive says the same of its late Fable, operator, Connes and moonshine material (p. 4): it is admitted only as a candidate interface, pending an explicit typed map and compatibility check.

## 8 What remains open, and what was not checked

### 8.1 Open questions

1. *The global obligation.* Show that for every prime  $p \equiv 1, 121, 169, 289, 361, 529 \pmod{840}$  some shell is occupied. This is the conjecture itself, restated by Theorem 3.1. The archive names it as the remaining obligation (p. 4). By Proposition 3.6, any proof must produce, for some  $R$ , a prime factor of  $(p + R)/4$  that is a non-residue modulo  $R$ , and must then go further, since the converse fails.
2. *The census survivors.* 2.394% of the final hard domain survives the carrier chain (§4.3). Can further carriers force them, or does a residue obstruction of Jacobi type block a finite chain? The archive makes no claim either way.
3. *The strict-record deficit-box conjecture* (§4.1; archive §19.2). It holds at the two strict records with non-empty deficit below 8,803,369 (reproduced here), and strict recordness cannot be dropped from it ( $p = 806,521$ ). A further test needs a strict record above 8,803,369, if one exists (Proposition 4.2).
4. *Growth of the least class.* Along the strict records up to 8,803,369,  $h$  takes the values 1, 2, 3, 6, 8, 15, 27 (Proposition 4.2). The workbench states no conjecture on the growth of  $h(p)$ , and none is stated here.

### 8.2 What this reader did not check

- *The archive.* Apart from the parts marked proved or reproduced in §2.2, its 619 pages were mapped, not audited. In particular, the side constructions (§§2, 4, 6–8, 9 after its opening, 21–23, the operator part of §24, and §25) and the literature audits (§§11–14 and 17) were not compared with the papers they audit.
- *Lean.* The Lean files (84 in the source archive) were not rebuilt. The archive’s own account of what each Lean module checks is quoted where it matters (§4.3).
- *The census.* The enumerations of §4.3 were not re-run. Only the arithmetic consistency of the reported counts was checked.
- *The supplement.* The general proofs of the supplement’s Sections 1–7 were not re-derived, except that of the no-two-shears theorem of Section 5 (Proposition 2.1); their finite content was reproduced in part (§2.3). Section 8, the auxiliary torus, was not checked.
- *The 124-page reader.* Only Proposition 2.2 was checked.
- *The continuation.* Items 1, 5, 9–13, 15–20, 24, 25, 27 and 28 were located, not checked. So were item 4’s enumeration, the deduction in item 23, and ES-04 to ES-08 beyond the ES-08 witness.
- *Novelty.* No literature search for novelty was made here. The workbench claims no novelty for the classical coordinates and identities it uses.

## A Catalogue of statements

This catalogue lists every statement of the workbench that this reader states, proves, reproduces or locates. For each it gives where the statement is in the workbench and how far it was checked here, in the labels defined in “About this reader”. “WB” quotes the workbench’s own label where it adds something. Locators are as in the front matter: “archive” is the 619-page PDF, “supplement” the 86-page supplement, and “item  $n$ ” the  $n$ -th entry of the README’s “Start reading” list.

| Statement                                                                                                                         | Where in the workbench                                          | Status here                                                                      | Here       |
|-----------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|----------------------------------------------------------------------------------|------------|
| <i>The shell reduction</i>                                                                                                        |                                                                 |                                                                                  |            |
| The ordered pairs completing a first denominator $a$ number $2 E_a  +  M_a $ ; solvable iff some shell is occupied (ES-01, ES-02) | supplement §1; archive §9, p. 38                                | Proved here, generalised to every odd prime and every $a > p/4$ with $p \nmid a$ | Thm 3.1    |
| The least denominator lies in $(p/4, p/2)$                                                                                        | a reduction step of item 22                                     | Proved here, for every odd prime                                                 | Thm 3.1(a) |
| A counterexample prime has $E_a = M_a = \emptyset$ at every shell (the all-shell no-hit sieve)                                    | supplement §1                                                   | Audited (43a_ §1.2); it is Thm 3.1(c)                                            | §3         |
| The residual-3 shell: criterion and count                                                                                         | supplement §1; archive §10                                      | Proved here; count corrected for $p \equiv 5 \pmod{12}$                          | Prop. 3.2  |
| The residual-7 shell: criterion                                                                                                   | supplement §1; archive §10                                      | Proved here, for every $p \equiv 1 \pmod{4}$                                     | Prop. 3.3  |
| Both first shells empty only if $p \equiv 1, 25, 121 \pmod{168}$                                                                  | first-pass observation                                          | Proved here                                                                      | Prop. 3.4  |
| Reduction to the six hard classes modulo 840                                                                                      | archive §10, p. 95 (classical)                                  | Proved here                                                                      | Prop. 3.5  |
| Jacobi-symbol barrier, and the failure of its converse                                                                            | archive §10, p. 96                                              | Proved here                                                                      | Prop. 3.6  |
| <i>Least classes, records and deficits</i>                                                                                        |                                                                 |                                                                                  |            |
| $h(p) = (R + 1)/4$ for the least occupied shell                                                                                   | archive §§17.4, 19                                              | Proved here                                                                      | Lemma 4.1  |
| The seven strict records up to 8,803,369; $1201 \in C_6 \setminus C_5$ ; $h(806521) = 15$                                         | archive §19.2, p. 152; §17.4, p. 132                            | Reproduced                                                                       | Prop. 4.2  |
| At $p_* = 8,803,369$ the least occupied shell has $R = 107$ , and it carries one unordered solution                               | archive §3                                                      | Reproduced                                                                       | §4.1       |
| The mod-107 deficit progression, the covering, and the two middle fibres (WB: credited to u/CommonCareful3149)                    | archive §16; bench R06                                          | Reproduced                                                                       | §4.1       |
| The deficits at 118,801 and 8,803,369; the boundary example 806,521                                                               | archive §19.3                                                   | Reproduced                                                                       | §4.1       |
| The strict-record deficit-box conjecture                                                                                          | archive §19.2 (WB: conjecture)                                  | Open                                                                             | §4.1, §8   |
| The shell $a = 3075$ at $p = 12289$ : counts (3, 6, 3)                                                                            | archive §20                                                     | Counts reproduced; the orbit and the quotient certificate located                | §2.2       |
| <i>Families and the census</i>                                                                                                    |                                                                 |                                                                                  |            |
| Four families solving the equation on $248k + 209, 248k + 89, 17112k + 11137, 17112k + 14113$                                     | archive §24, p. 423                                             | Proved here; that they meet every hard class is new here                         | Thm 4.3    |
| The carrier census: 97.606% of the final hard domain forced                                                                       | archive pp. 3–5 and §24 (WB: bounded sufficient-carrier census) | Located; the arithmetic of the counts checked                                    | §4.3       |
| The residual-eleven support ideal; the paired $R = 7/R = 11$ affine theorem                                                       | archive §18                                                     | Located                                                                          | §2.2       |

| Statement                                                                                                                                          | Where in the workbench                             | Status here                                                                               | Here      |
|----------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|-------------------------------------------------------------------------------------------|-----------|
| <i>The supplement and the 124-page reader</i>                                                                                                      |                                                    |                                                                                           |           |
| The finite content of Sections 1–7 (gluing, ratio transport, shears, successors, orchard partition, boundary law, unary box, incidence, raw scale) | supplement §§1–7                                   | Reproduced in part                                                                        | §2.3      |
| No two shear edges compose                                                                                                                         | supplement §5 (stated for $p \equiv 1 \pmod{12}$ ) | Generalised here to $p \not\equiv 2 \pmod{3}$ ; fails for $p \equiv 2 \pmod{3}$           | Prop. 2.1 |
| The Navier–Stokes auxiliary cover                                                                                                                  | supplement §8                                      | Not checked                                                                               | §2.3      |
| The quartic of a solution: $p = -5u_4/u_3$ and the coefficient relation                                                                            | 124-page reader; item 11                           | Proved here (also 21_)                                                                    | Prop. 2.2 |
| The Fable cover and the weighted conductor                                                                                                         | 124-page reader                                    | Located                                                                                   | §2.4      |
| <i>The September continuation</i>                                                                                                                  |                                                    |                                                                                           |           |
| Item 2: a seven-cycle sink at $p = 2521$ (WB: method counterexample)                                                                               | research/incoming/                                 | The sink audited (certificate); the closure theorems located                              | §5, §6    |
| Item 3: a closed triangle at a 22-digit prime                                                                                                      | research/incoming/                                 | The triangle audited (certificate), its primality proved here; the escape theorem located | §5        |
| Item 4: six nonresidue primes within 25 factorisations (WB: computer-assisted)                                                                     | research/incoming/                                 | Reductions read; conditional on an enumeration not re-run                                 | §5        |
| Item 6: the principal-only bound is negative in sum                                                                                                | research/incoming/                                 | Audited                                                                                   | §5        |
| Item 7: least-nonresidue localisation                                                                                                              | research/incoming/                                 | Audited                                                                                   | §5        |
| Item 8: the middle identity and its bounds                                                                                                         | research/incoming/                                 | Reproduced on 1,613 middle states; the atlas located                                      | §5        |
| Items 14 and 21: template bound; same-word returns                                                                                                 | research/incoming/                                 | Audited                                                                                   | §5        |
| Item 22: the degree-8 invariant is negative on solutions                                                                                           | research/incoming/                                 | Audited (certificate)                                                                     | §5        |
| Items 23 and 26                                                                                                                                    | research/incoming/                                 | Audited in part                                                                           | §5        |
| Items 1, 5, 9–13, 15–20, 24, 25, 27, 28                                                                                                            | research/incoming/                                 | Located (item 11’s quartic proved here)                                                   | 43a_ §2   |
| ES-09: the $(3, 4, \infty)$ monodromy covers                                                                                                       | received/es_s6_counterfactual/                     | Audited for odd levels; false for even levels                                             | §5.2      |
| ES-08: a forced solution at $p = 1201$ , $a = 310$ , $R = 39$                                                                                      | 12 September continuation                          | The witness checked; the forcing inequality not audited                                   | 43a_ §4   |
| Results bench R01–R10, arguments X1–X7                                                                                                             | RESULTS.md                                         | Checked in 43a_ §3 (X1 not recomputed)                                                    | §5        |
| <i>The rest of the archive</i>                                                                                                                     |                                                    |                                                                                           |           |
| Side constructions: §§2, 4–8, 21–23, 25 and the operator part of §24                                                                               | archive                                            | Located; §5 read; the fibre of §7 verified in the zeta reader                             | §2.2, §7  |
| Literature audits: §§11–14 and 17                                                                                                                  | archive                                            | Located; the classes of §17 replayed                                                      | §2.2      |

| Statement                                                  | Where in the workbench | Status here | Here |
|------------------------------------------------------------|------------------------|-------------|------|
| The Star–Kneser forcing theorem and its retracted converse | archive §15            | Located     | §6   |

B How this reader was checked

Sources.

- The repository KokunoYumeto/erdos-strauss-foundation at commit d20b32e, read-only.
- The 619-page archive and its source archive, downloaded on 26 September 2026 from the Zenodo record 10.5281/zenodo.22678971:

00\_ERDOS\_STRAUSS\_Project\_Reader.pdfMD5 c1ee553967e0e6f12d4976e171c11b1501\_ERDOS\_STRAUSS\_Source\_Verification\_and\_Machine\_Memory.zipMD5 4129ec50d0c02ff7daf61fde6c589de4

The main source file of the archive (60,561 lines) and its fourteen satellite files were read where they are cited.
- The supplement’s source, in the repository folder research/expanded-2026-09-08/exact\_bounded\_transport\_72/.

**What was refereed earlier.** Theorem 3.1 and Propositions 3.2–3.5, the audited items of Section 5, and Section 5.2 come from the Erdős–Straus section of the reader of four workbenches. There they went through five stages:

1. a first-pass audit by a separate Claude instance, which read the workbench and wrote independent checks (43a\_);
2. a referee pass on the notes 43\_–47\_, by an instance that had written none of them; it found, among other things, that the orbit statements of Section 5.2 fail at even levels;
3. a referee pass on that reader;
4. two further passes that read it for understatement and missed generality as well as for overstatement.

These passes produced the range  $x < p/2$  in Theorem 3.1, its extension to all odd primes, Propositions 3.4 and 3.5, and the corrected count of Proposition 3.2 at  $p \equiv 5 \pmod{12}$ . The checks are checks/workbenches/workbench\_reader\_claims\_checks.py and checks/workbenches/generality\_checks.py; the referee reports are in checks/workbenches/.

**What is new in this reader, and has been checked only by its author.**

- Proposition 3.6;
- Lemma 4.1 and Proposition 4.2, with the reproduction of the deficit data in §4.1;
- Theorem 4.3, and the statement that each family meets every hard class;
- Proposition 2.1;
- Proposition 2.2, whose relation was also checked in 21\_;
- the map of the archive (§2.2), the consistency of the census counts (§4.3), and the three new rows of the overlap table (Section 7).

**The scripts of this reader.** They import no workbench code and recompute every number from the definitions. They are in `checks/es_reader/` on the branch named below.

| Script                           | What it checks                                                                                                                                                                                                                  | Time, result   |
|----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| <code>esr_records.py</code>      | $h(p)$ for every prime up to 8,803,369 through Lemma 4.1 and Theorem 3.1; the strict records; $h(806521) = h(118801) = 15$                                                                                                      | 3 s; as stated |
| <code>esr_lemma_checks.py</code> | Proposition 3.6 on all 47,210 shells meeting its hypothesis at odd primes below 3000 (none occupied), and the failure of its converse; Lemma 4.1 against a brute-force enumeration of all solutions at the odd primes below 700 | 25 s; all pass |
| <code>esr_deficits.py</code>     | the deficit data of §4.1 at 73, 1129, 1201, 21169, 118,801, 8,803,369 and 806,521, their independence of the primitive root, the corollary of archive §16, and the counts $ E_a  = 0$ , $ M_a  = 2$ at $p_*$ (22 checks)        | 1 s; all pass  |
| <code>esr_families.py</code>     | the four identities of Theorem 4.3 as polynomial identities, exact fractions for $k \leq 200$ , the hard classes met and the least prime in each                                                                                | 1 s; all pass  |
| <code>esr_twoshears.py</code>    | Proposition 2.1: the full shear graph of every prime below 2000, and a complete search for two-edge paths up to $10^7$                                                                                                          | 23 s; all pass |

**A partial second pass.** A second-pass audit of the supplement was started with a separate Claude instance and was stopped by a usage limit before it wrote its report. Its scripts and outputs are kept in `checks/es_reader/second_pass_partial/`, and §2.3 reports what they show. The one reported mismatch was traced to the script’s own expectation (§2.3).

**What has not been done.** This version has not had an independent referee pass of its own. The earlier readers of this audit each had at least one, by a Claude instance that had written none of the text; the most recent of these passes read for understatement and missed generality as well as for overstatement. Such instances were unavailable when this version was written, because of a usage limit at the time. A referee pass of the same kind is planned, and its findings will be applied before this reader is deposited with the workbench’s Zenodo collection. Until then, the statements listed above as new rest on this author’s proofs and scripts alone.

**Where everything is.** This reader, its sources and its scripts are on the branch `claude/claude-ab-grind-20260925` of `KokunoYumeto/zeta-function-research-reader`, in `contrib/claude-ab/grind_20260925/`: the reader in `es_reader/`, the scripts in `checks/es_reader/`, and the audit notes `21_`, `43_`, `43a_`, `47_` and `47a_` at the top level.

## References

- [1] *A compact complex threefold fibred by tori over the projective line, and the six-sphere*, manuscript circulated by L. Alpöge, 2026, <https://alpo.ge/s6.pdf>.
- [2] C. Elsholtz, T. Tao, *Counting the number of solutions to the Erdős–Straus equation on unit fractions*, J. Aust. Math. Soc. **94** (2013), 50–105, doi:10.1017/S1446788712000468.
- [3] E. J. Ionascu, A. Wilson, *On the Erdős–Straus conjecture*, arXiv:1001.1100 (2010).
- [4] Spiridon Mihnea, Dumitru C. Bogdan (author names as listed on arXiv), *Further verification and empirical evidence for the Erdős–Straus conjecture*, arXiv:2509.00128 (2025).
- [5] L. J. Mordell, *Diophantine Equations*, Academic Press, London, 1969.
- [6] S. E. Salez, *The Erdős–Straus conjecture: new modular equations and checking up to  $N = 10^{17}$* , arXiv:1406.6307 (2014).

- [7] R. C. Vaughan, *On a problem of Erdős, Straus and Schinzel*, *Mathematika* (1970), doi:10.1112/S0025579300002886.
- [8] Claude (Anthropic), *The split-zero programme around the Riemann zeta function: a reader of its results, with proofs*, version 2, Zenodo, 2026, DOI 10.5281/zenodo.22977243 (version 1: DOI 10.5281/zenodo.22970703; Lemma 6.6 has the same number in both).
- [9] Claude (Anthropic), *Four workbenches and where they meet: a reader of the Erdős–Straus, Collatz, Erdős Problem 817 and Yang–Mills workbenches*, 2026, file 00-claude-opus-5-5-workbench-reader.pdf of the Zenodo record DOI 10.5281/zenodo.22977243; source on the branch claude/claude-ab-grind-20260925 of KokunoYumeto/zeta-function-research-reader, folder contrib/claude-ab/grind\_20260925/workbench\_reader/.